Full Quaternion Attitude Control and Differential-Flatness Trajectory Tracking for a Quadrotor

Project Report - Group 12 (B Sainath CB.SC.U4AIE24309, Vishal CB.SC.U4AIE24363) - School of Artificial Intelligence, Amrita Vishwa Vidyapeetham. Base papers: Fresk & Nikolakopoulos (ECC 2013) [1] and Choutri, Lagha, Dala & Lipatov (IEEE 2017) [2]. All equations below are typed out and were cross-checked numerically against the implementation (7 equation groups, randomised trials - see Section 8).

1. Introduction

A quadrotor is a rigid body with four rotors. It has six degrees of freedom in attitude and position but only four independent control inputs: the total rotor thrust and the three body torques. This underactuation is the central difficulty of quadrotor control. The vehicle cannot accelerate sideways directly; it must first tilt so that the thrust vector, which always points along the body z-axis, points partly in the desired direction. Attitude and translation are therefore coupled, and any trajectory-tracking controller must respect that coupling rather than fight it.

This project implements, verifies and extends two papers that treat the two halves of this problem. The first, Fresk and Nikolakopoulos [1], is a pure attitude controller: a nonlinear proportional-squared quaternion law that commands body torques directly from the quaternion attitude error. The second, Choutri, Lagha, Dala and Lipatov [2], adds trajectory tracking: it uses differential flatness to convert a desired path into the thrust and attitude that produce it, and closes both loops with linear quadratic regulators. The project reproduces both papers from their equations, flies the results on four simulation stacks (an ideal model, MuJoCo contact physics, gym-pybullet-drones, and real ArduPilot firmware in software-in-the-loop), and extends the second paper with a trajectory library. Every result quoted in this report regenerates from a single command, and the repository carries 85 tests of which 83 pass in the canonical environment (two require simulator stacks not installed there).

> **How the two papers connect**
>
> Paper 1 controls attitude only - point the body, and accept wherever the resulting thrust pushes the vehicle. Paper 2 inverts this: given where the vehicle should be, compute the thrust and the attitude that put it there, then let an attitude loop hold that attitude. Our implementation keeps both on one airframe (0.2 kg, inertia diag(6.5e-4, 6.5e-4, 1.2e-3) kg m^2), so the comparison between them is a property of the controllers, not of different vehicles.

2. Quaternion representation of attitude

An attitude is a rotation of the body frame relative to the world frame. Euler angles represent it with three numbers but contain a singularity (gimbal lock at pitch of plus or minus 90 degrees); the rotation matrix avoids that singularity but carries nine constrained numbers. A unit quaternion carries four numbers with one norm constraint, has no singularity anywhere, and composes rotations with a cheap product. Both papers base their controllers on it. Throughout, quaternions are written scalar-first, q = (q0, q1, q2, q3) with vector part q-vector = (q1, q2, q3), and represent the rotation from body frame to world frame.

2.1 Algebra

```
q = q0 + q1*i + q2*j + q3*k = (q0, q-vector),   q0 in R, q-vector in R^3
```

A quaternion: one scalar part and a three-vector part. A unit quaternion (norm 1) represents a rotation.

```
p (x) q = ( p0*q0 - p-vector . q-vector ,

p0*q-vector + q0*p-vector + p-vector x q-vector )
```

The quaternion product. The dot and cross products of the vector parts appear explicitly; the product is associative but not commutative, which is exactly what makes it suitable for composing rotations in order.

```
q* = (q0, -q-vector)

||q|| = sqrt( q (x) q* ) = sqrt(q0^2 + q1^2 + q2^2 + q3^2)

q^-1 = q* / ||q||      ( = q* for a unit quaternion )
```

Conjugate, norm and inverse. Conjugation reverses a rotation; for unit quaternions the conjugate is the inverse.

```
v' = q (x) [0, v] (x) q*      which equals  R(q) * v
```

Rotating a vector v by the attitude q. R(q) is the 3x3 direction cosine matrix; its columns are the body axes written in world coordinates, it is orthonormal, and R(p (x) q) = R(p) R(q).

```
q = [ cos(alpha/2),  u * sin(alpha/2) ]
```

Axis-angle form: a rotation by angle alpha about unit axis u. This is the most useful construction rule - it builds reference quaternions directly from geometric intent.

2.2 The kinematic equation

```
q-dot = (1/2) * q (x) [0, omega]        omega = body angular rate

q-dot = (1/2) * [0, omega_w] (x) q      omega_w = R(q) omega (world)
```

The two equivalent forms of the quaternion kinematic equation. One physical motion, two algebraic shapes: match the form to the frame your angular velocity lives in. Mixing them produces an error that grows with tilt times rate - small near hover, radical during aggressive manoeuvres (Section 7, finding on validation).

> **Why quaternions in a control loop**
>
> The quaternion error between a reference and a measurement is itself a quaternion, its vector part points along the axis that rotates one onto the other, and its magnitude tracks the error angle - a controller can act on it directly with no trigonometry and no singularity. Euler angles would need unwrapping and would fail at gimbal lock; the paper's flip test rotates through exactly the region where Euler representations misbehave.

3. Paper 1: full quaternion based attitude control

3.1 The plant

The attitude dynamics of a rigid body about its centre of mass, with the paper's inertia tensor and input torque, are:

```
J * omega-dot = tau ,      J = diag( 6.5e-4, 6.5e-4, 1.2e-3 )  kg*m^2
```

The rotational plant: torque divided by inertia gives angular acceleration. The paper's vehicle has equal roll and pitch inertia and roughly double the yaw inertia, as expected for a planar-X quadrotor with light arms.

Coupled with the kinematic equation of Section 2.2, this is a double integrator from torque to attitude: torque shapes the angular rate, the rate shapes the attitude. That structure is what makes a derivative-free proportional law sufficient - no integral action is needed for the tracking

error to decay.

3.2 The control law

The controller [1] acts entirely in quaternion space. Given the reference attitude q_ref and the measured attitude q_m (both in the paper's conjugated convention), the error quaternion, its vector part, and the torque command are:

```
      q_err  = q_ref (x) q_m*                          (attitude error)
       e      = vector part of q_err, sign-corrected so it takes the
            SHORTEST rotation (flip sign when q_err's scalar is negative)
         tau    = -Pq * e  -  Pw * omega_m ,      Pq = 20,  Pw = 4
              tau    clipped elementwise to +/- 4 N*m per axis
```

The complete law. The first term rotates the attitude error to zero; the second damps the angular rate. Torque to attitude is a double integrator, so this proportional-derivative structure in quaternion form drives the error to zero without an integral term. The gains and the saturation bound are the paper's stated values and are reproduced exactly. The measured attitude and rate carry the paper's sensor noise: uniform plus or minus 0.1 on every quaternion component and rate component, refreshed at 1 kHz.

Why the sign correction matters: q and -q represent the same rotation, so an error quaternion with a negative scalar would steer the long way around. Correcting the sign of the vector part makes the controller always take the short rotation - except in the flip scenario, where the correction is deliberately disabled so the vehicle commits to the full 360-degree rotation instead of unwinding.

3.3 The three benchmark scenarios

- STEP - unit-radian (57.3 degree) steps on roll, pitch and yaw, staggered at t = 1, 5 and 9 s across a 15 s run. Essentially no overshoot; about 2 s settling per axis.
- SINE - 0.5 radian amplitude, 1 rad/s sine on all three axes, phase-shifted so the torques stay out of phase. Measured lag of 0.4 to 0.5 s matches the paper's stated about half a second.
- FLIP - a full 0 to 2*pi rad roll ramp over 2 s. The raw quaternion components stay smooth through the whole rotation with no gimbal-lock artifact, which is the paper's central claim.

3.4 A derived result: the digital control-rate bound

The paper specifies no digital control rate. Linearising the closed loop about identity attitude gives a characteristic equation and its poles:

```
                  I*e-double-dot + Pw*e-dot + (Pq/2)*e = 0
          lambda(1,2) = ( -Pw +/- sqrt(Pw^2 - 2*I*Pq) ) / (2*I)
                              =  -2.5  and  -6150  rad/s
```

The continuous loop is heavily overdamped and perfectly stable. The instability that appears at modest control rates is purely a sampling artifact of the fast pole: at a 200 Hz update the discrete closed-loop spectral radius is 29.9 (unstable); at 1 kHz it is 5.16; at 10 kHz it is just under one.

```
f_control >= Pw / (m * I_min)  with margin m = 0.5   gives   f_control >=
                                  12.3 kHz
```

The derived, not tuned, minimum safe control rate. Our simulator adopts it by default and the scheduler refuses to run below it; the instability at 200 Hz can be reproduced directly by overriding the rate.

4. Paper 2: differential flatness and trajectory tracking

4.1 The full model

Paper 2 extends the attitude plant to the complete six-degree-of-freedom vehicle. The state is position p, velocity v, attitude quaternion q and body rate omega; the inputs are the total thrust T and the body torque tau:

```
                    p-dot  = v

          v-dot  = R(q) * [0, 0, T/m]  -  [0, 0, g]

              q-dot  = (1/2) * q (x) [0, omega]

          J*omega-dot = - omega x (J*omega) + tau
```

The four rows say: velocity is the derivative of position; the thrust vector (magnitude T along the body z-axis, rotated into the world by R(q)) accelerates the vehicle against gravity; the attitude evolves with the body rate; and the torque overcomes the gyroscopic term to change the rate. The second row contains the underactuation: the direction of the thrust is chosen by the attitude, so horizontal acceleration is bought with tilt.

The four rotor speeds produce the two inputs through a mixing matrix. Each rotor i contributes thrust proportional to the square of its speed, b times omega-i-squared, along body z, plus a reaction torque about body z whose sign alternates with the propeller direction. For the X-configuration with arm distance d:

```
          [T ]   [  b     b     b     b  ] [w1]

          [tx] = [  b*d   b*d  -b*d  -b*d ] [w2]

          [ty]   [ -b*d   b*d   b*d  -b*d ] [w3]

          [tz]   [ -c     c    -c     c  ] [w4]

   so [T, tx, ty, tz] = M [w1, w2, w3, w4],  w = M^-1 [T, tx, ty, tz]
```

The mixer: front rotors 1 and 2 (columns 1-2) differ from rear rotors 3 and 4 for the roll torque tx; left differs from right for ty; the alternating signs of the reaction coefficient c give the yaw torque tz. Inverting M turns any thrust-and-torque command into rotor speeds; every real flight controller contains this matrix. Rotor speeds are bounded, so when a command exceeds what the rotors can produce, we deliver the thrust exactly and scale the torque uniformly to the largest feasible value - naive clipping instead corrupts the total thrust and destabilises altitude (a finding of this project).

Linearising the model at hover (identity attitude, zero rates) shows how tilt steers: the horizontal acceleration becomes proportional to the small quaternion tilt components, and the thrust commands the vertical acceleration:

```
      v-dot approximately = [ 2*(T/m)*q2 ,  -(T/m)*q1 ,  T/m - g ]

                q-vector-dot approximately = omega / 2
```

The hover linearisation. These two relations are the seeds of the two linear models the LQR design uses.

4.2 Differential flatness

A system is differentially flat if its complete state and input can be written as functions of a set of flat outputs and finitely many of their derivatives - with no integration of the differential equations. For the quadrotor the flat outputs are the position and the yaw angle:

```
                    sigma(t) = ( x, y, z, psi )
```

Four flat outputs for four inputs. Given these and their derivatives up to second order, every state and input of the vehicle follows algebraically.

The chain that recovers the vehicle from the outputs starts from Newton's law on the translational row: if the vehicle is to accelerate with a, the net force must be m*a, and the only forces are

thrust and gravity, so the thrust vector is determined:

```
        f  = m * ( a + g*z-hat )          the required thrust vector
    T_d = ||f||                        its magnitude (the thrust command)
              d = f / ||f||                      its direction
        R(q_d) * z-hat = d                body z must point along d
      yaw of q_d = psi_d                the remaining attitude freedom
```

The direction d fixes two degrees of the three-degree attitude; the chosen yaw fixes the last one. The attitude reference q_d is therefore fully determined by the desired acceleration and yaw - this is the inversion of the underactuation described in Section 1. Differentiating q_d in time would additionally give the reference body rate and, one derivative further, the reference torque (the full parameterisation; our inner loop regulates rate to zero instead, a documented simplification).

The paper constructs q_d as a composition of two quaternions, and we implement that construction literally:

```
                q_pd = tilt quaternion taking z-hat onto d:
       axis = z-hat x d  (horizontal, so q_pd's yaw element is exactly 0)
        q_pd = [ cos(alpha/2), sin(alpha/2) * (z-hat x d)/||z-hat x d|| ]
                    with alpha the angle between z-hat and d
    q_zd = [ cos(psi_d/2), 0, 0, sin(psi_d/2) ]     pure yaw quaternion
      q_d  = q_pd (x) q_zd                            the composition
```

Eq. (19), (20), (21) of the paper, typed as implemented. The tilt quaternion's rotation axis is horizontal by construction, so its yaw component is identically zero - the paper's statement that the position quaternion's fourth element vanishes. We verified numerically that the composed q_d aligns the body z-axis with the thrust direction to 1e-10 and that the tilt part's yaw element is zero to 1e-12. The only degeneracy of the whole chain is a vanishing thrust vector (free fall), where the attitude stops mattering to the motion.

4.3 The double-loop LQR

The controller is cascaded the way the plant is: an outer position loop computes the acceleration command, the flatness map of the previous subsection turns it into the thrust T_d and the attitude reference q_d, and an inner attitude loop tracks q_d with torque. The inner loop must run roughly an order of magnitude faster than the outer - the classical time-scale separation. Both loops are linear quadratic regulators, designed on the hover linearisation:

```
              INNER loop, state x = [q-vector(3), omega(3)]:
       x-dot = A x + B u,   A = [ 0(3x3)  (1/2)I ]    B = [   0    ]
                                [ 0(3x3)   0    ]        [ J^-1   ]
              OUTER loop, per axis, state x = [p, p-dot]:
         x-dot = A x + B u,   A = [ 0  1 ]              B = [ 0 ]
                                  [ 0  0 ]                  [ 1 ]
```

The inner A-matrix's half-identity block is the hover relation q-vector-dot = omega/2; the B-matrix's J-inverse is Euler's equation. The outer loop is the double integrator with acceleration as input. Both have the standard linear form x-dot = A x + B u.

```
J_cost = (1/2) * integral from 0 to infinity of ( x' Q x + u' R u ) dt

P A + A' P - P B R^-1 B' P + Q = 0        (the Riccati equation)

K = R^-1 B' P ,        u = -K x
```

The LQR design: the quadratic cost trades state error (weighted by Q) against control effort (weighted by R); the algebraic Riccati equation yields the matrix P, and the constant gain K multiplies the state to give the optimal input. For the outer double integrator the resulting law is a proportional-derivative controller with optimal gains plus the flatness feedforward. The paper states the controller ran at 200 Hz; we run both loops at 200 Hz with zero-order hold, and the closed-loop poles of both designs are verified Hurwitz (all eigenvalues with negative real part) in our tests.

4.4 The study case and results

The paper's experiment tracks a circular path at a fixed altitude of 1 m, starting from the point (-1, 0, 0), with the 200 Hz controller and actuator limits in effect. We reproduce it with the vehicle starting at rest and climbing onto the circle with a minimum-jerk altitude ramp, which reproduces the roughly one-second lag the paper reports in the horizontal responses. The root-mean-square tracking errors after settling, on three plants:

Plant	RMS x	RMS y	RMS z	RMS radial	Peak rotor
Ideal Eq. (8) model, RK4 at 1 kHz	1.6 cm	3.6 cm	0.6 cm	2.7 cm	0.57 N
MuJoCo contact physics, 20 kHz	1.4 cm	3.7 cm	0.7 cm	2.8 cm	0.99 N
MuJoCo + paper noise +/-0.1	2.3 cm	4.6 cm	0.9 cm	3.7 cm	1.69 N

Approximately 97 percent radial accuracy on the 1 m circle, with no rotor saturation (the per-rotor limit is 1.95 N), and near-identical numbers on two independent plants - the flatness reference is exact regardless of the integrator behind it. Where the paper gives no numbers (the LQR weights, the circle speed, the actuator lag model, the gravity value, the integrator), our choices are documented in the repository README's assumptions table.

5. Cross-simulator deployment of the attitude law

The paper 1 controller, with gains untouched, was deployed through a shared bridge to four stacks: MuJoCo, gym-pybullet-drones, Gazebo under WSL2, and real ArduPilot firmware compiled from source under WSL2. All four run the same references and the same noise model. The sine scenario is the clean comparison:

Stack	Sine RMS attitude error	Flip outcome
ArduPilot firmware (real code)	14.7 deg +/- 0.2	completes, settles 2.89 s
Gazebo (real physics)	17.3 deg +/- 0.0	engine-dependent knife-edge
gym-pybullet-drones	17.52 deg	knife-edge regime
MuJoCo	17.61 deg	completes, settles 2.93 s

The four stacks agree to within three degrees on sine tracking, and the two native stacks agree to 0.03 degrees - evidence that the loop belongs to the controller rather than to any engine. The residual error near 17.5 degrees is the sensor noise floor: the paper's uniform plus-or-minus 0.1 quaternion-component noise corresponds to roughly 11 degrees of measured attitude error, which the controller cannot remove because it cannot see through it. The flip's engine dependence is analysed in the repository as a knife-edge property of the pursuit equilibrium: with shortest-path disabled the vehicle rate saturates at five times the sine of half the lag angle, so a 2-second ramp is trackable only with a lag of about 1.36 rad, and whether the vehicle crosses the full rotation depends on integrator fidelity at exactly that margin.

6. Extensions implemented (trajectory library)

Beyond the two papers we extended the flatness stack with a trajectory library - the same frozen controllers, new flat-output generators only. The waypoint trajectories are piecewise degree-7 polynomials that minimise the integral of squared snap (the fourth derivative), because flatness makes the control input depend on the derivatives: jerk drives body rates and snap drives torques, so the smoothest flyable path is the minimum-snap path. Waypoints, rest-to-rest ends and velocity continuity at the joints are constraints; the minimum is found per segment in closed form. Tracking results, all with the frozen LQR gains:

Scenario	RMS x	RMS y	RMS z	RMS 3D	Peak rotor
circle (paper 2 study case)	1.6	3.6	0.6	3.9	0.57
figure-8, ideal model	2.5	3.9	0.9	4.7	0.57
figure-8, MuJoCo contact physics	2.4	4.0	1.0	4.7	1.01
vertical orbit (cone sweep)	1.6	3.5	0.6	3.9	0.57
waypoint tour, minimum-snap	4.6	3.9	1.3	6.1	0.83
waypoint tour, minimum-jerk	5.8	5.2	2.3	8.1	0.84

(all values in centimetres; rotor peaks in newtons against the 1.95 N limit). The minimum-snap tour tracks 25 percent better than the minimum-jerk tour through identical waypoints at identical times - the flatness argument (snap drives torques) measured directly. Building it surfaced a instructive failure: minimising only snap with position-only waypoints lets long segments loiter and then teleport, because position is four integrals away from the cost and slow large excursions are nearly free. Pinning the corridor tangent velocity at each waypoint removes the freedom; solving in normalised time per segment removes a numerical conditioning problem that reproduces the same pathology even with correct constraints.

7. Verification methodology

Every claim in this report is backed by an automated check. The repository carries 85 tests: quaternion algebra and convention consistency, plant behaviour, mixer round-trips and desaturation, LQR pole stability, closed-loop tracking thresholds on both plants, trajectory waypoint accuracy and knot continuity. The paper 2 implementation was additionally reviewed equation-by-equation against the paper by three independent reviewers, with every finding re-verified numerically before being acted on. That review confirmed 17 findings, none high-severity, of which two were real defects that we fixed:

- The plant had paired body-frame rates with the world-frame form of the quaternion kinematic equation. The two forms coincide at hover but diverge with tilt times rate; on a 2-second, 2 rad/s spin the wrong pairing accumulated 1.5 rad of attitude error while the correct one stayed at 3e-8. The plant now uses the literal body-rate form.
- The attitude reference was built by a Gram-Schmidt construction whose comment claimed exact equivalence to the paper's composition; measurement showed a twist about the thrust axis growing quadratically with tilt. It was replaced with the literal composition of Section 4.2, whose tilt component's yaw element is zero to machine precision.

For this report, the equations typed in Sections 2 to 4 were cross-checked once more against the implementation in seven randomised groups - quaternion algebra and rotation matrices (50 trials), the paper 1 control law including its saturation stage (200 trials), the plant inertia, all four rows of the paper 2 model, the mixer matrix and its inverse, the flatness map including the zero yaw element of the tilt quaternion (30 trials), and the two state-space models with their Riccati gains verified Hurwitz. All seven groups pass. The recordings distributed with the project are the

measured runs themselves: the recording hooks attach to the benchmark loops, and each recording's metrics match the corresponding benchmark to the last digit.

8. Conclusions

- The quaternion attitude law of paper 1 was reproduced exactly (gains, saturation, noise), extended to a six-degree-of-freedom flight stack, and flown unchanged on four simulation stacks including real ArduPilot firmware, with cross-engine agreement of 0.03 degrees RMS on the native stacks.
- A derived digital-stability bound - a minimum control rate of 12.3 kHz for these gains - explains why naive discrete implementations of the paper's controller diverge, and is enforced by our scheduler.
- The differential-flatness tracker of paper 2 was implemented equation-by-equation, validated by independent review with all findings numerically confirmed, and tracks the paper's study case to 2.7 to 2.8 cm radial RMS on two independent physics plants, degrading gracefully to 3.7 cm under the paper's full sensor noise.
- The extension trajectory library (figure-8, vertical orbit, minimum-snap waypoint tours) runs on the same frozen controllers and quantifies the flatness argument: minimum-snap tracks 25 percent better than minimum-jerk on identical waypoints.
- Everything regenerates from single commands; 83 of 85 tests pass in the canonical environment and the two skips require simulator stacks not installed there.

9. References

[1] E. Fresk and G. Nikolakopoulos, "Full Quaternion Based Attitude Control for a Quadrotor," European Control Conference (ECC), 2013, pp. 3864-3869.

[2] K. Choutri, M. Lagha, L. Dala and M. Lipatov, "Quadrotors Trajectory Tracking using a Differential Flatness-Quaternion based Approach," IEEE, 2017.

[3] D. Mellinger and V. Kumar, "Minimum Snap Trajectory Generation and Control for Quadrotors," ICRA, 2011. (basis of the extension trajectory library)

[4] T. Lee, M. Leok and N. H. McClamroch, "Geometric Tracking Control of a Quadrotor UAV on SE(3)," CDC, 2010. (comparison target for the planned third controller family)